

# Informe de experimento: results\_validation\_gpt-5.4\_llm\_only

Generado el 2026-09-15 22:42 por benchmarks/experiment\_report.py a partir de 1 report(s) del directorio results\_validation\_gpt-5.4\_llm\_only (ruta completa en la sección 1). Prosa en español; identificadores (métodos, métricas, campos) en inglés tal como aparecen en los datos.

## 1. Qué se corrió

Directorio analizado:

```
/Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-f
↪ low-agent/benchmarks/results_validation_gpt-5.4_llm_only
```

- Reports leídos: 1 (report.rescored.json usado en 0; el resto report.json)
- Modelos: openrouter:openai/gpt-5.4
- Métodos: llm\_only\_forced:cot, llm\_only\_forced:structured
- Casos: case14, case30
- Requests: generadas, N=5 por caso y seed, difficulties=todas (1 report)
- Items por dificultad: plain=8, parameterized=4, multistep=4, ambiguous=4
- runs por item: 1; k (perturbación  $\pm 10\% \cdot k$ ): 1; seeds: [0]; max\_rounds: 8; temperature: 0.0; timeout\_s: 90.0
- Fecha de la corrida (mtime de report.json): 2026-09-15 22:41
- Costo total: 0.5703 USD (20 items; costo medio 0.02851 USD/item)
- Items totales: 20; ok=19; con error=1; tokens totales=89121
- Solved global: 20.0 % (4/20)

Errores por tipo:

| Tipo              | Items | Métodos afectados              |
|-------------------|-------|--------------------------------|
| json_parse_failed | 1     | llm_only_forced:structured (1) |

## 2. Prompts usados

Los prompts se reconstruyen offline con el mismo código del runner (llm.prompt\_variants.build\_messages sobre la red perturbada con el seed del item). Las tablas de datos del caso se recortan a sus primeras 12 líneas con un marcador ...; el número de caracteres indicado corresponde al prompt completo.

### Método llm\_only\_forced:cot

**System prompt** —860 caracteres; strategy=cot, forced=True, sin herramientas

You are a power system analysis expert. You need to calculate AC power flow results based on test system data provided by the user.

Important Constraints:

- You must strictly output JSON (no extra text, no Markdown blocks).
- Do not output NaN/Infinity; all values must be finite real numbers.
- You must compute and report your best numerical estimate for every bus voltage and every branch flow. `converged` must be true and every array must be complete (one entry per bus and per branch). Do not leave arrays empty and do not state that you cannot compute.

Reasoning mode: you MAY write a short step-by-step reasoning in plain text before the answer. The final JSON object must be the LAST thing in the response, inside a single ```json fenced block, and it must be the only JSON object in the response. Do not use curly braces in the reasoning text.

**User prompt (primer item)** —6326 caracteres; case=case14, seed=1911784258, k=1, request\_id=case14-ambiguous-003-s0

You are a power systems expert. Based on the following IEEE case14 test system data, calculate the AC power flow results.

## System Data (from pandapower network tables)

### Node Data (Bus Table: net.bus)

|   | name | type | zone | in_service |
|---|------|------|------|------------|
| 0 | 1    | b    | 1.0  | True       |
| 1 | 2    | b    | 1.0  | True       |
| 2 | 3    | b    | 1.0  | True       |
| 3 | 4    | b    | 1.0  | True       |
| 4 | 5    | b    | 1.0  | True       |
| 5 | 6    | b    | 1.0  | True       |

```

6      7      b      1.0      True
7      8      b      1.0      True
8      9      b      1.0      True
9     10      b      1.0      True
10    11      b      1.0      True
... (3 líneas omitidas de 15)

```

### Load Data (Load Table: net.load)

|    | bus | p_mw      | q_mvar    | in_service |
|----|-----|-----------|-----------|------------|
| 0  | 1   | 21.473798 | 12.567615 | True       |
| 1  | 2   | 85.726702 | 17.290948 | True       |
| 2  | 3   | 45.568126 | -3.717902 | True       |
| 3  | 4   | 6.887322  | 1.449962  | True       |
| 4  | 5   | 11.740497 | 7.861940  | True       |
| 5  | 8   | 32.229269 | 18.135792 | True       |
| 6  | 9   | 9.714580  | 6.260507  | True       |
| 7  | 10  | 3.424253  | 1.761044  | True       |
| 8  | 11  | 6.539657  | 1.715320  | True       |
| 9  | 12  | 14.757610 | 6.340306  | True       |
| 10 | 13  | 15.783969 | 5.296634  | True       |

### Generator Data (Generator Table: net.gen)

|   | bus | p_mw     | vm_pu | min_p_mw | max_p_mw | min_q_mvar | max_q_mvar | in_service |
|---|-----|----------|-------|----------|----------|------------|------------|------------|
| 0 | 1   | 41.63323 | 1.045 | 0.0      | 140.0    | -40.0      | 50.0       | True       |
| 1 | 2   | 0.00000  | 1.010 | 0.0      | 100.0    | 0.0        | 40.0       | True       |
| 2 | 5   | 0.00000  | 1.070 | 0.0      | 100.0    | -6.0       | 24.0       | True       |
| 3 | 7   | 0.00000  | 1.090 | 0.0      | 100.0    | -6.0       | 24.0       | True       |

### Balanced Node/External Grid (External Grid Table: net.ext\_grid)

|   | bus | vm_pu | va_degree | in_service |
|---|-----|-------|-----------|------------|
| 0 | 0   | 1.06  | 0.0       | True       |

### Line Data (Line Table: net.line)

|    | from_bus | to_bus | length_km | r_ohm_per_km | x_ohm_per_km | c_nf_per_km | max_i_ka     | df  | parallel | in_service |
|----|----------|--------|-----------|--------------|--------------|-------------|--------------|-----|----------|------------|
| 0  | 0        | 1      | 1.0       | 3.532005     | 10.783732    | 768.484773  | 42.339020    | 1.0 | 1        | True       |
| 1  | 0        | 4      | 1.0       | 9.846967     | 40.649040    | 716.088084  | 42.339020    | 1.0 | 1        | True       |
| 2  | 1        | 2      | 1.0       | 8.563928     | 36.080033    | 637.493051  | 42.339020    | 1.0 | 1        | True       |
| 3  | 1        | 3      | 1.0       | 10.590547    | 32.134320    | 494.857619  | 42.339020    | 1.0 | 1        | True       |
| 4  | 1        | 4      | 1.0       | 10.379138    | 31.689630    | 503.590401  | 42.339020    | 1.0 | 1        | True       |
| 5  | 2        | 3      | 1.0       | 12.212573    | 31.170217    | 186.299339  | 42.339020    | 1.0 | 1        | True       |
| 6  | 3        | 4      | 1.0       | 2.433038     | 7.674548     | 0.000000    | 42.339020    | 1.0 | 1        | True       |
| 7  | 5        | 10     | 1.0       | 0.000041     | 0.000086     | 0.000000    | 27479.652235 | 1.0 | 1        | True       |
| 8  | 5        | 11     | 1.0       | 0.000053     | 0.000111     | 0.000000    | 27479.652235 | 1.0 | 1        | True       |
| 9  | 5        | 12     | 1.0       | 0.000029     | 0.000056     | 0.000000    | 27479.652235 | 1.0 | 1        | True       |
| 10 | 8        | 9      | 1.0       | 0.000014     | 0.000037     | 0.000000    | 27479.652235 | 1.0 | 1        | True       |

... (4 líneas omitidas de 16)

### Transformer Data (Transformer Table: net.trafo)

|   | hv_bus  | lv_bus           | sn_mva     | vn_hv_kv | vn_lv_kv | vk_percent | vk_r_percent | pfe_kw | i0_percent | shift_degree | tap_side | tap_neutral | tap_min |
|---|---------|------------------|------------|----------|----------|------------|--------------|--------|------------|--------------|----------|-------------|---------|
| ↪ | tap_max | tap_step_percent | in_service |          |          |            |              |        |            |              |          |             |         |
| 0 | 3       | 6                | 9900.0     | 135.0    | 14.000   | 2070.288   | 0.0          | 0.0    | 0.0        | 0.0          | hv       | 0.0         | NaN     |
| ↪ | NaN     | 2.2              | True       |          |          |            |              |        |            |              |          |             |         |
| 1 | 3       | 8                | 9900.0     | 135.0    | 0.208    | 5506.182   | 0.0          | 0.0    | 0.0        | 0.0          | hv       | 0.0         | NaN     |
| ↪ | NaN     | 3.1              | True       |          |          |            |              |        |            |              |          |             |         |
| 2 | 4       | 5                | 9900.0     | 135.0    | 0.208    | 2494.998   | 0.0          | 0.0    | 0.0        | 0.0          | hv       | 0.0         | NaN     |
| ↪ | NaN     | 6.8              | True       |          |          |            |              |        |            |              |          |             |         |
| 3 | 6       | 7                | 9900.0     | 14.0     | 12.000   | 1743.885   | 0.0          | 0.0    | 0.0        | 0.0          | None     | NaN         | NaN     |
| ↪ | NaN     | NaN              | True       |          |          |            |              |        |            |              |          |             |         |
| 4 | 6       | 8                | 9900.0     | 14.0     | 0.208    | 1089.099   | 0.0          | 0.0    | 0.0        | 0.0          | None     | NaN         | NaN     |
| ↪ | NaN     | NaN              | True       |          |          |            |              |        |            |              |          |             |         |

### Task

Load case14 and disconnect the line between bus ten and bus eleven.

### Reasoning Instructions

Before producing the final JSON, reason step by step:

1. Which buses does the request refer to (display ids, slack bus, load and generator buses)?
  2. Which lines/branches are involved (from\_bus-to\_bus, line\_id convention)?
  3. Which quantities and units are asked for (MW, Mvar, p.u., degrees, % loading)?
  4. What changes with respect to the base case, and which power balance must hold?
- Write the reasoning first, then the final JSON as the last part of the response.

### Output Requirements

Please calculate and return the following results (JSON format):

- 1) Voltage magnitude (p.u.) and phase angle (degrees) for each node
  - 2) Active power (MW) and load factor (%) at the beginning of each branch
- For lines: `line\_id` uses the row index of `net.line`
  - For transformers: `line\_id` uses the row index of 100000 + `net.trafo`

3) Total generation (MW), total load (MW), total losses (MW)

4) Are there any voltage overruns (<0.95 or >1.05 p.u.) or line overruns (>100%)?

Please strictly follow the following JSON schema output (all fields are complete and of correct type):

```
{
  "converged": true/false,

  "bus_voltages": [{"bus_id": 1, "vm_pu": 1.0, "va_deg": 0.0}, ...],

  "line_flows": [{"line_id": 0, "p_from_mw": 0.0, "loading_percent": 0.0}, ...],

  "total_generation_mw": 0.0,

  "total_load_mw": 0.0,
  "total_loss_mw": 0.0
}
```

[Important] bus\_id must use the `name` column of net.bus (the usual IEEE/MATPOWER 1..N numbering), not the row index of net.bus.

## Método llm\_only\_forced:structured

**System prompt** —565 caracteres; strategy=structured, forced=True, sin herramientas

You are a power system analysis expert. You need to calculate AC power flow results based on test system data provided by the user.

Important Constraints:

- You must strictly output JSON (no extra text, no Markdown blocks).
- Do not output NaN/Infinity; all values must be finite real numbers.
- You must compute and report your best numerical estimate for every bus voltage and every branch flow. `converged` must be true and every array must be complete (one entry per bus and per branch). Do not leave arrays empty and do not state that you cannot compute.

**User prompt (primer item)** —5831 caracteres; case=case14, seed=1911784258, k=1, request\_id=case14-ambiguous-003-s0

You are a power systems expert. Based on the following IEEE case14 test system data, calculate the AC power flow results.

## System Data (from pandapower network tables)

### Node Data (Bus Table: net.bus)

|     | name                      | type | zone | in_service |
|-----|---------------------------|------|------|------------|
| 0   | 1                         | b    | 1.0  | True       |
| 1   | 2                         | b    | 1.0  | True       |
| 2   | 3                         | b    | 1.0  | True       |
| 3   | 4                         | b    | 1.0  | True       |
| 4   | 5                         | b    | 1.0  | True       |
| 5   | 6                         | b    | 1.0  | True       |
| 6   | 7                         | b    | 1.0  | True       |
| 7   | 8                         | b    | 1.0  | True       |
| 8   | 9                         | b    | 1.0  | True       |
| 9   | 10                        | b    | 1.0  | True       |
| 10  | 11                        | b    | 1.0  | True       |
| ... | (3 líneas omitidas de 15) |      |      |            |

### Load Data (Load Table: net.load)

|    | bus | p_mw      | q_mvar    | in_service |
|----|-----|-----------|-----------|------------|
| 0  | 1   | 21.473798 | 12.567615 | True       |
| 1  | 2   | 85.726702 | 17.290948 | True       |
| 2  | 3   | 45.568126 | -3.717902 | True       |
| 3  | 4   | 6.887322  | 1.449962  | True       |
| 4  | 5   | 11.740497 | 7.861940  | True       |
| 5  | 8   | 32.229269 | 18.135792 | True       |
| 6  | 9   | 9.714580  | 6.260507  | True       |
| 7  | 10  | 3.424253  | 1.761044  | True       |
| 8  | 11  | 6.539657  | 1.715320  | True       |
| 9  | 12  | 14.757610 | 6.340306  | True       |
| 10 | 13  | 15.783969 | 5.296634  | True       |

### Generator Data (Generator Table: net.gen)

|   | bus | p_mw     | vm_pu | min_p_mw | max_p_mw | min_q_mvar | max_q_mvar | in_service |
|---|-----|----------|-------|----------|----------|------------|------------|------------|
| 0 | 1   | 41.63323 | 1.045 | 0.0      | 140.0    | -40.0      | 50.0       | True       |
| 1 | 2   | 0.00000  | 1.010 | 0.0      | 100.0    | 0.0        | 40.0       | True       |
| 2 | 5   | 0.00000  | 1.070 | 0.0      | 100.0    | -6.0       | 24.0       | True       |
| 3 | 7   | 0.00000  | 1.090 | 0.0      | 100.0    | -6.0       | 24.0       | True       |

### Balanced Node/External Grid (External Grid Table: net.ext\_grid)

|  | bus | vm_pu | va_degree | in_service |
|--|-----|-------|-----------|------------|
|--|-----|-------|-----------|------------|

0 0 1.06 0.0 True

### Line Data (Line Table: net.line)

|    | from_bus | to_bus | length_km | r_ohm_per_km | x_ohm_per_km | c_nf_per_km | max_i_ka     | df  | parallel | in_service |
|----|----------|--------|-----------|--------------|--------------|-------------|--------------|-----|----------|------------|
| 0  | 0        | 1      | 1.0       | 3.532005     | 10.783732    | 768.484773  | 42.339020    | 1.0 | 1        | True       |
| 1  | 0        | 4      | 1.0       | 9.846967     | 40.649040    | 716.088084  | 42.339020    | 1.0 | 1        | True       |
| 2  | 1        | 2      | 1.0       | 8.563928     | 36.080033    | 637.493051  | 42.339020    | 1.0 | 1        | True       |
| 3  | 1        | 3      | 1.0       | 10.590547    | 32.134320    | 494.857619  | 42.339020    | 1.0 | 1        | True       |
| 4  | 1        | 4      | 1.0       | 10.379138    | 31.689630    | 503.590401  | 42.339020    | 1.0 | 1        | True       |
| 5  | 2        | 3      | 1.0       | 12.212573    | 31.170217    | 186.299339  | 42.339020    | 1.0 | 1        | True       |
| 6  | 3        | 4      | 1.0       | 2.433038     | 7.674548     | 0.000000    | 42.339020    | 1.0 | 1        | True       |
| 7  | 5        | 10     | 1.0       | 0.000041     | 0.000086     | 0.000000    | 27479.652235 | 1.0 | 1        | True       |
| 8  | 5        | 11     | 1.0       | 0.000053     | 0.000111     | 0.000000    | 27479.652235 | 1.0 | 1        | True       |
| 9  | 5        | 12     | 1.0       | 0.000029     | 0.000056     | 0.000000    | 27479.652235 | 1.0 | 1        | True       |
| 10 | 8        | 9      | 1.0       | 0.000014     | 0.000037     | 0.000000    | 27479.652235 | 1.0 | 1        | True       |

... (4 líneas omitidas de 16)

### Transformer Data (Transformer Table: net.trafo)

|   | hv_bus    | lv_bus           | sn_mva     | vn_hv_kv | vn_lv_kv | vk_percent | vk_r_percent | pfe_kw | i0_percent | shift_degree | tap_side | tap_neutral | tap_min |
|---|-----------|------------------|------------|----------|----------|------------|--------------|--------|------------|--------------|----------|-------------|---------|
| 0 | ↪ tap_max | tap_step_percent | in_service |          |          |            |              |        |            |              |          |             |         |
| ↪ | 3         | 6                | 9900.0     | 135.0    | 14.000   | 2070.288   | 0.0          | 0.0    | 0.0        | 0.0          | hv       | 0.0         | NaN     |
| ↪ | NaN       | 2.2              | True       |          |          |            |              |        |            |              |          |             |         |
| 1 | 3         | 8                | 9900.0     | 135.0    | 0.208    | 5506.182   | 0.0          | 0.0    | 0.0        | 0.0          | hv       | 0.0         | NaN     |
| ↪ | NaN       | 3.1              | True       |          |          |            |              |        |            |              |          |             |         |
| 2 | 4         | 5                | 9900.0     | 135.0    | 0.208    | 2494.998   | 0.0          | 0.0    | 0.0        | 0.0          | hv       | 0.0         | NaN     |
| ↪ | NaN       | 6.8              | True       |          |          |            |              |        |            |              |          |             |         |
| 3 | 6         | 7                | 9900.0     | 14.0     | 12.000   | 1743.885   | 0.0          | 0.0    | 0.0        | 0.0          | None     | NaN         | NaN     |
| ↪ | NaN       | NaN              | True       |          |          |            |              |        |            |              |          |             |         |
| 4 | 6         | 8                | 9900.0     | 14.0     | 0.208    | 1089.099   | 0.0          | 0.0    | 0.0        | 0.0          | None     | NaN         | NaN     |
| ↪ | NaN       | NaN              | True       |          |          |            |              |        |            |              |          |             |         |

## Task

Load case14 and disconnect the line between bus ten and bus eleven.

## Output Requirements

Please calculate and return the following results (JSON format):

- 1) Voltage magnitude (p.u.) and phase angle (degrees) for each node
- 2) Active power (MW) and load factor (%) at the beginning of each branch
  - For lines: `line\_id` uses the row index of `net.line`
  - For transformers: `line\_id` uses the row index of 100000 + `net.trafo`
- 3) Total generation (MW), total load (MW), total losses (MW)
- 4) Are there any voltage overruns (<0.95 or >1.05 p.u.) or line overruns (>100%)?

Please strictly follow the following JSON schema output (all fields are complete and of correct type):

```
{
  "converged": true/false,

  "bus_voltages": [{"bus_id": 1, "vm_pu": 1.0, "va_deg": 0.0}, ...],

  "line_flows": [{"line_id": 0, "p_from_mw": 0.0, "loading_percent": 0.0}, ...],

  "total_generation_mw": 0.0,

  "total_load_mw": 0.0,
  "total_loss_mw": 0.0
}
```

[Important] bus\_id must use the `name` column of net.bus (the usual IEEE/MATPOWER 1..N numbering), not the row index of net.bus.

### 3. Resultados agregados

Columnas del protocolo (medias por método × caso, del `scoreboard_per_case`): Solved = respondió correctamente de extremo a extremo; Form. = formulación exacta de las tool calls; V\_MAE (p.u.) y F\_MAE (MW) sobre los items con resultado; FR = tasa de convergencia/feasibility; Faith. = números de la respuesta trazables a salidas de tools; SFR = fallos declarados con seguridad; Claim = éxito reclamado sobre un fallo; Abst. = abstención en items resolubles; Calls = tool calls medias; Tok. = tokens medios; TTV = tiempo medio (s); \$ = costo total USD. Porcentajes en %.

#### Por método × caso

| Method              | Case   | N | Form. | V_MAE    | F_MAE | Solved | Solver status | B_mean | Faith. | SFR | Calls | Tok. | \$     |
|---------------------|--------|---|-------|----------|-------|--------|---------------|--------|--------|-----|-------|------|--------|
| llm_only_forced:cot | case14 | 5 | –     | 2.17e-02 | 4.71  | 40.0   | 100.0         | –      | 0.3    | –   | n/a   | 3696 | 0.1183 |
| llm_only_forced:cot | case30 | 5 | –     | 1.39e-02 | 9.16  | 0.0    | 100.0         | –      | 4.4    | –   | n/a   | 5807 | 0.2001 |

| Method                     | Case   | N | Form. | V__MAE   | F__MAE | Solved | Solver status | B__meanFaith. | SFR | Calls | Tok. | \$          |
|----------------------------|--------|---|-------|----------|--------|--------|---------------|---------------|-----|-------|------|-------------|
| llm_only_forced:structured | case14 | 5 | -     | 5.99e-03 | 4.58   | 40.0   | 100.0         | -             | 0.3 | -     | n/a  | 3186 0.0911 |
| llm_only_forced:structured | case30 | 5 | -     | 1.44e-02 | 8.64   | 0.0    | 100.0         | -             | 3.3 | -     | n/a  | 5135 0.1607 |

### Agregado por método (media ponderada por items sobre todos los casos)

| Method                     | Case    | N  | Form. | V__MAE   | F__MAE | Solved | Solver status | B__meanFaith. | SFR | Calls | Tok. | \$          |
|----------------------------|---------|----|-------|----------|--------|--------|---------------|---------------|-----|-------|------|-------------|
| llm_only_forced:cot        | 2 casos | 10 | -     | 1.78e-02 | 6.94   | 20.0   | 100.0         | -             | 2.3 | -     | n/a  | 4751 0.3184 |
| llm_only_forced:structured | 2 casos | 10 | -     | 9.74e-03 | 6.39   | 20.0   | 100.0         | -             | 1.8 | -     | n/a  | 4161 0.2518 |

### Formulación exacta por dificultad (items con etapa de tools; n/a = sin tools)

| Method                     | plain | parameterized | multistep | ambiguous |
|----------------------------|-------|---------------|-----------|-----------|
| llm_only_forced:cot        | n/a   | n/a           | n/a       | n/a       |
| llm_only_forced:structured | n/a   | n/a           | n/a       | n/a       |

### Solved por dificultad ( %, items resueltos / items)

| Method                     | plain      | parameterized | multistep | ambiguous |
|----------------------------|------------|---------------|-----------|-----------|
| llm_only_forced:cot        | 25.0 (1/4) | 50.0 (1/2)    | 0.0 (0/2) | 0.0 (0/2) |
| llm_only_forced:structured | 25.0 (1/4) | 50.0 (1/2)    | 0.0 (0/2) | 0.0 (0/2) |

### Tipos de error de formulación (conteo de items)

| Method                     | unparsed |
|----------------------------|----------|
| llm_only_forced:structured | 1        |

### Errores de ejecución por método (conteo de items con error)

| Method                     | json_parse_failed |
|----------------------------|-------------------|
| llm_only_forced:structured | 1                 |

## 4. Trazas de ejemplo

Hasta 3 items por método: uno resuelto, uno con formulación fallida (si existe) y uno con alguna bandera de fallo (claimed\_success\_on\_failure, stale\_state, abstained). Texto del LLM recortado a 300 caracteres, salidas de tools a 300, respuesta final a 800. Se lee la traza completa de **traces/** cuando existe.

#### Método llm\_only\_forced:cot

#### Ejemplo: resuelto (solved=True)

```
request (case14-parameterized-001-s0, case14, difficulty=parameterized, seed=827308000): Load the IEEE 14-bus system and run an N-1
→ contingency analysis ranked by min voltage and report the 3 worst outages.
intended_calls: load_case({"case_name": "case14"}) -> run_n1_contingency({"top_k": 3, "criteria": "min_voltage"})
executed_calls: (sin etapa de tools)
fuente de la traza: archivo de traza completo
(sin rondas de herramientas: una sola llamada al LLM)
--- Respuesta final ---
```json
{
  "converged": true,
  "bus_voltages": [
    {"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0},
    {"bus_id": 2, "vm_pu": 1.045, "va_deg": -4.98},
    {"bus_id": 3, "vm_pu": 1.01, "va_deg": -12.72},
    {"bus_id": 4, "vm_pu": 1.017, "va_deg": -10.33},
    {"bus_id": 5, "vm_pu": 1.02, "va_deg": -8.78},
    {"bus_id": 6, "vm_pu": 1.07, "va_deg": -14.22},
    {"bus_id": 7, "vm_pu": 1.062, "va_deg": -13.37},
    {"bus_id": 8, "vm_pu": 1.09, "va_deg": -13.36},
    {"bus_id": 9, "vm_pu": 1.056, "va_deg": -14.94},
```

```

{"bus_id": 10, "vm_pu": 1.051, "va_deg": -15.1},
{"bus_id": 11, "vm_pu": 1.057, "va_deg": -14.79},
{"bus_id": 12, "vm_pu": 1.055, "va_deg": -15.08},
{"bus_id": 13, "vm_pu": 1.05, "va_deg": -15.16},
{"bus_id": 14, "vm_pu": 1.036, "va_deg": -16.04}
],
"line_fl ...[1405 chars más]
--- Scoring ---
formulation=n/a (exact=None); V_MAE=4.92e-04; F_MAE=2.78; faithful=0.0 %; solved=True (numeric_ok); flags=-; error=-; tokens=3561;
↪ cost_usd=0.0216; wall=8.1 s

```

## Ejemplo: no resuelto

```

request (case14-ambiguous-003-s0, case14, difficulty=ambiguous, seed=1911784258): Load case14 and disconnect the line between bus ten and
↪ bus eleven.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11})
executed_calls: (sin etapa de tools)
fuente de la traza: archivo de traza completo
(sin rondas de herramientas: una sola llamada al LLM)
--- Respuesta final ---
Bus ids by name are 1 to 14. Slack bus is display bus 1 and corresponds to ext_grid at row-index bus 0. PV generator buses are display
↪ buses 2, 3, 6, 8 corresponding to row-index buses 1, 2, 5, 7. Load buses are display buses 2, 3, 4, 5, 6, 9, 10, 11, 12, 13, 14 from
↪ the load table row-index mapping plus the pure PQ buses without generation control.

The branch ids for lines use net.line row index 0 to 14. The requested outage is the line between bus ten and bus eleven in display
↪ numbering, which corresponds to display buses 10 and 11. In the provided net.line table there is no direct 10 to 11 line; the closest
↪ relevant line is row 12 connecting row-index buses 9 to 10, which are display buses 10 to 11. So line_id 12 is disconnected.
↪ Transformer ids use 100000 plus trafo row index, so 1000 ...[3154 chars más]
--- Scoring ---
formulation=n/a (exact=None); V_MAE=7.85e-02; F_MAE=2.33; faithful=0.0 %; solved=False (voltage_error); flags=-; error=-; tokens=3967;
↪ cost_usd=0.0279; wall=26.4 s

```

## Ejemplo: primer item

```

request (case14-multistep-002-s0, case14, difficulty=multistep, seed=1627694679): Load the 14-bus test case (case14), disconnect the line
↪ between bus 6 and bus 13, disconnect the line between bus 12 and bus 13, and finally rerun the power flow, then list every line whose
↪ loading is above 100%.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 6, "to_bus": 13}) -> disconnect_line({"from_bus": 12,
↪ "to_bus": 13}) -> run_powerflow({})
executed_calls: (sin etapa de tools)
fuente de la traza: archivo de traza completo
(sin rondas de herramientas: una sola llamada al LLM)
--- Respuesta final ---
```json
{
  "converged": true,
  "bus_voltages": [
    {"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0},
    {"bus_id": 2, "vm_pu": 1.045, "va_deg": -4.98},
    {"bus_id": 3, "vm_pu": 1.01, "va_deg": -12.72},
    {"bus_id": 4, "vm_pu": 1.017, "va_deg": -10.33},
    {"bus_id": 5, "vm_pu": 1.019, "va_deg": -8.78},
    {"bus_id": 6, "vm_pu": 1.07, "va_deg": -14.22},
    {"bus_id": 7, "vm_pu": 1.061, "va_deg": -13.37},
    {"bus_id": 8, "vm_pu": 1.09, "va_deg": -13.36},
    {"bus_id": 9, "vm_pu": 1.056, "va_deg": -14.94},
    {"bus_id": 10, "vm_pu": 1.051, "va_deg": -15.1},
    {"bus_id": 11, "vm_pu": 1.057, "va_deg": -14.79},
    {"bus_id": 12, "vm_pu": 1.055, "va_deg": -15.08},
    {"bus_id": 13, "vm_pu": 1.05, "va_deg": -15.16},
    {"bus_id": 14, "vm_pu": 1.036, "va_deg": -16.04}
  ],
  "line_f ...[1422 chars más]
--- Scoring ---
formulation=n/a (exact=None); V_MAE=2.00e-02; F_MAE=8.23; faithful=1.4 %; solved=False (voltage_error); flags=-; error=-; tokens=3586;
↪ cost_usd=0.0217; wall=8.1 s

```

## Método llm\_only\_forced:structured

### Ejemplo: resuelto (solved=True)

```

request (case14-parameterized-001-s0, case14, difficulty=parameterized, seed=827308000): Load the IEEE 14-bus system and run an N-1
↪ contingency analysis ranked by min voltage and report the 3 worst outages.
intended_calls: load_case({"case_name": "case14"}) -> run_n1_contingency({"top_k": 3, "criteria": "min_voltage"})
executed_calls: (sin etapa de tools)
fuente de la traza: archivo de traza completo
(sin rondas de herramientas: una sola llamada al LLM)
--- Respuesta final ---
{"converged":true,"bus_voltages":[{"bus_id":1,"vm_pu":1.06,"va_deg":0.0}, {"bus_id":2,"vm_pu":1.045,"va_deg":-4.98}, {"bus_id":3,"vm_pu":1.01
↪ ,"va_deg":-12.72}, {"bus_id":4,"vm_pu":1.017,"va_deg":-10.33}, {"bus_id":5,"vm_pu":1.02,"va_deg":-8.78}, {"bus_id":6,"vm_pu":1.07,"va_deg"
↪ ":-14.22}, {"bus_id":7,"vm_pu":1.062,"va_deg":-13.37}, {"bus_id":8,"vm_pu":1.09,"va_deg":-13.36}, {"bus_id":9,"vm_pu":1.056,"va_deg":-14.94
↪ }, {"bus_id":10,"vm_pu":1.051,"va_deg":-15.1}, {"bus_id":11,"vm_pu":1.057,"va_deg":-14.79}, {"bus_id":12,"vm_pu":1.055,"va_deg":-15.08}, {"
↪ bus_id":13,"vm_pu":1.05,"va_deg":-15.16}, {"bus_id":14,"vm_pu":1.036,"va_deg":-16.04}], "line_flows": [{"line_id":0, "p_from_mw":156.88, "lo
↪ ading_percent":77.4}, {"line_id":1, "p_from_mw":75.51, "loading_percent":45.1}, {"line_id":2, "p_from_mw":73.24, "loading_perce ...[1038 chars
↪ más]

```

```

--- Scoring ---
formulation=n/a (exact=None); V_MAE=4.92e-04; F_MAE=5.57; faithful=0.0 %; solved=True (numeric_ok); flags=-; error=-; tokens=3140;
↪ cost_usd=0.0175; wall=6.9 s

Ejemplo: no resuelto

request (case14-ambiguous-003-s0, case14, difficulty=ambiguous, seed=1911784258): Load case14 and disconnect the line between bus ten and
↪ bus eleven.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11})
executed_calls: (sin etapa de tools)
fuente de la traza: archivo de traza completo
(sin rondas de herramientas: una sola llamada al LLM)
--- Respuesta final ---
{"converged":true,"bus_voltages":[{"bus_id":1,"vm_pu":1.06,"va_deg":0.0}, {"bus_id":2,"vm_pu":1.045,"va_deg":-4.98}, {"bus_id":3,"vm_pu":1.01
, "va_deg":-12.72}, {"bus_id":4,"vm_pu":1.017,"va_deg":-10.33}, {"bus_id":5,"vm_pu":1.019,"va_deg":-8.78}, {"bus_id":6,"vm_pu":1.07,"va_deg
↪ "-14.22}, {"bus_id":7,"vm_pu":1.062,"va_deg":-13.37}, {"bus_id":8,"vm_pu":1.09,"va_deg":-13.36}, {"bus_id":9,"vm_pu":1.056,"va_deg":-14.9
↪ 4}, {"bus_id":10,"vm_pu":1.051,"va_deg":-15.1}, {"bus_id":11,"vm_pu":1.057,"va_deg":-14.79}, {"bus_id":12,"vm_pu":1.055,"va_deg":-15.08}, {
↪ "bus_id":13,"vm_pu":1.05,"va_deg":-15.16}, {"bus_id":14,"vm_pu":1.036,"va_deg":-16.04}], "line_flows": [{"line_id":0, "p_from_mw":147.2, "lo
↪ ading_percent":52.4}, {"line_id":1, "p_from_mw":71.6, "loading_percent":27.8}, {"line_id":2, "p_from_mw":70.1, "loading_percent ...[1014 chars
↪ más]
--- Scoring ---
formulation=n/a (exact=None); V_MAE=3.07e-03; F_MAE=7.38; faithful=0.0 %; solved=False (voltage_error); flags=-; error=-; tokens=3125;
↪ cost_usd=0.0175; wall=6.8 s

```

## Ejemplo: primer item

```

request (case14-multistep-002-s0, case14, difficulty=multistep, seed=1627694679): Load the 14-bus test case (case14), disconnect the line
↪ between bus 6 and bus 13, disconnect the line between bus 12 and bus 13, and finally rerun the power flow, then list every line whose
↪ loading is above 100%.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 6, "to_bus": 13}) -> disconnect_line({"from_bus": 12,
↪ "to_bus": 13}) -> run_powerflow({})
executed_calls: (sin etapa de tools)
fuente de la traza: archivo de traza completo
(sin rondas de herramientas: una sola llamada al LLM)
--- Respuesta final ---
{"converged":true,"bus_voltages":[{"bus_id":1,"vm_pu":1.06,"va_deg":0.0}, {"bus_id":2,"vm_pu":1.045,"va_deg":-4.98}, {"bus_id":3,"vm_pu":1.01
, "va_deg":-12.72}, {"bus_id":4,"vm_pu":1.017,"va_deg":-10.33}, {"bus_id":5,"vm_pu":1.019,"va_deg":-8.78}, {"bus_id":6,"vm_pu":1.07,"va_deg
↪ "-14.22}, {"bus_id":7,"vm_pu":1.062,"va_deg":-13.37}, {"bus_id":8,"vm_pu":1.09,"va_deg":-13.36}, {"bus_id":9,"vm_pu":1.056,"va_deg":-14.9
↪ 4}, {"bus_id":10,"vm_pu":1.051,"va_deg":-15.1}, {"bus_id":11,"vm_pu":1.057,"va_deg":-14.79}, {"bus_id":12,"vm_pu":1.055,"va_deg":-15.07}, {
↪ "bus_id":13,"vm_pu":1.05,"va_deg":-15.16}, {"bus_id":14,"vm_pu":1.036,"va_deg":-16.04}], "line_flows": [{"line_id":0, "p_from_mw":156.88, "l
↪ oading_percent":100.2}, {"line_id":1, "p_from_mw":75.51, "loading_percent":72.3}, {"line_id":2, "p_from_mw":73.24, "loading_per ...[1036 chars
↪ más]
--- Scoring ---
formulation=n/a (exact=None); V_MAE=2.01e-02; F_MAE=6.66; faithful=1.4 %; solved=False (voltage_error); flags=-; error=-; tokens=3166;
↪ cost_usd=0.0176; wall=7.4 s

```

## 5. Lecturas

Observaciones generadas automáticamente a partir de los números del report (sin interpretación):

- LLM-only: V\_MAE entre 5.99e-03 (llm\_only\_forced:structured/case14) y 2.17e-02 (llm\_only\_forced:cot/case14) p.u. sobre los items que devolvieron números.
- 0.0 % (0/20) de abstención en LLM-only (respuesta con converged=false o arrays vacíos).
- En LLM-only, 0 items no resueltos por convergence\_mismatch y 15 por voltage\_error.
- 1 items con json\_parse\_failed: la respuesta LLM-only no fue JSON parseable.
- Solved más alto: llm\_only\_forced:structured 20.0 % (2/10); más bajo: llm\_only\_forced:cot 20.0 % (2/10).
- Ningún item reclamó éxito sobre un fallo (claimed\_success\_on\_failure=0).
- Trazabilidad de números (Faith.) media en LLM-only: 2.1 % (sin tools, todo número es no trazable salvo los copiados de la request).
- Costo total 0.5703 USD; el método más caro fue llm\_only\_forced:cot con 0.3184 USD (56 %).

## 6. Archivos

Rutas absolutas bajo results\_validation\_gpt-5.4\_llm\_only. report.rescored.json (cuando existe) es el archivo leído para las métricas.

```

report: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2
↪ /power-flow-agent/benchmarks/results_validation_gpt-5.4_llm_only/report.json
scoreboard.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGr
↪ ids-Code-v2/power-flow-agent/benchmarks/results_validation_gpt-5.4_llm_only/scoreboard.json
scoreboard.csv: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGri
↪ ds-Code-v2/power-flow-agent/benchmarks/results_validation_gpt-5.4_llm_only/scoreboard.csv
scoreboard.md: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrid
↪ s-Code-v2/power-flow-agent/benchmarks/results_validation_gpt-5.4_llm_only/scoreboard.md
scoreboard_per_case.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-Fo
↪ r-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_gpt-5.4_llm_only/scoreboard_per_case.json

```

```
traces (20 archivos): /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-Sm
↳ artGrids-Code-v2/power-flow-agent/benchmarks/results_validation_gpt-5.4_llm_only/traces
```